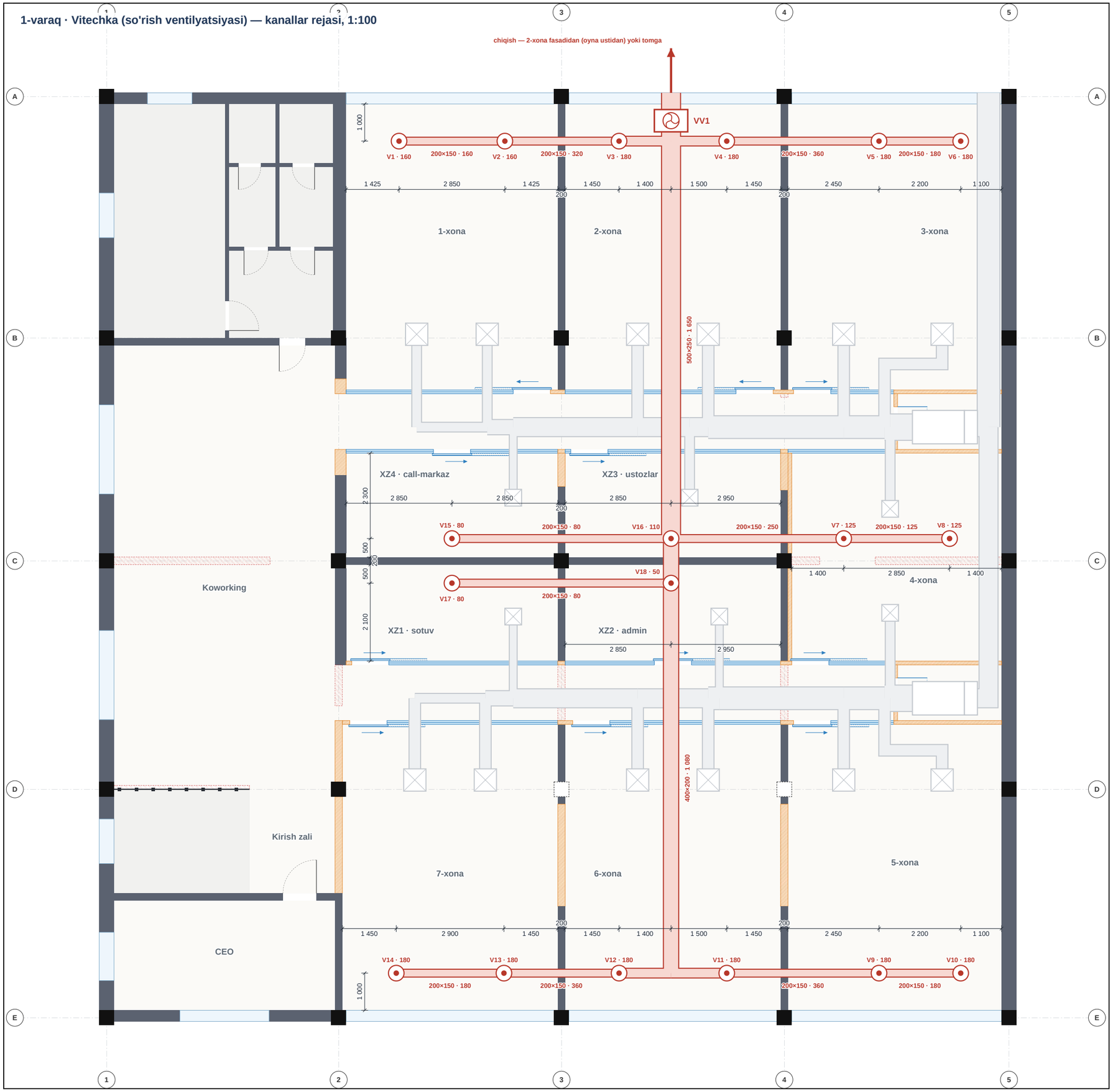
1-varaq · Vitechka (so'rish ventilyatsiyasi) — kanallar rejasi, 1:100



VITECHKA — SO'RISH KANALLARI (buyurtmachi eskizi bo'yicha)

so'rish kanali (kesim × havo, m³/soat)
so'rish panjarasi (klapan) V…
VV1 — kanalli ventilyator
konditsioner kanallari (2-varaq, xira)

Panjaralar — 18 ta, jami 2 690 m³/soat

Kod	Xona	x, mm	y, mm	m³/soat	Ø, mm
V1	1-xona	7 675	1 000	160	200
V2	1-xona	10 525	1 000	160	200
V3	2-xona	13 600	1 000	180	200
V4	2-xona	16 500	1 000	180	200
V5	3-xona	20 600	1 000	180	200
V6	3-xona	22 800	1 000	180	200
V7	4-xona	19 650	11 700	125	200
V8	4-xona	22 500	11 700	125	200
V9	5-xona	20 600	23 400	180	200
V10	5-xona	22 800	23 400	180	200
V11	6-xona	16 500	23 400	180	200
V12	6-xona	13 600	23 400	180	200
V13	7-xona	10 500	23 400	180	200
V14	7-xona	7 600	23 400	180	200
V15	XZ4 · call-markaz	9 100	11 700	80	200
V16	XZ3 · ustozlar	15 000	11 700	110	200
V17	XZ1 · sotuv	9 100	12 900	80	200
V18	XZ2 · admin	15 000	12 900	50	200

x — chap (g'arbiy) fasad ichki yuzidan, y — yuqori fasad ichki yuzidan (1-varaqdagi o'qlar bo'yicha). So'rish havosi — xonaga konditsioner orqali kelayotgan tashqi havoning (30%) 90% i: xonalar biroz ortiqcha bosimda, koridordan chang kirmaydi.

Kanallar (galvanizlangan po'lat, to'rtburchak, tekis yotqiziladi)

Liniya	m³/soat	Kesim, mm	Uzunlik, m
1-xona, 2-xona (g'arbiy), fasad bo'ylab	160–320	**200 × 150**	5.9
1-xona, 2-xona (g'arbiy), fasad bo'ylab	500	**250 × 150**	1.4
2-xona (sharqiy), 3-xona, fasad bo'ylab	180–360	**200 × 150**	6.3
2-xona (sharqiy), 3-xona, fasad bo'ylab	540	**250 × 150**	1.5
7-xona, 6-xona (g'arbiy), fasad bo'ylab	180–360	**200 × 150**	6.0
7-xona, 6-xona (g'arbiy), fasad bo'ylab	540	**250 × 150**	1.4
6-xona (sharqiy), 5-xona, fasad bo'ylab	180–360	**200 × 150**	6.3
6-xona (sharqiy), 5-xona, fasad bo'ylab	540	**250 × 150**	1.5
XZ4 (o'rta devor yonida)	80	**200 × 150**	5.9
4-xona (XZ3 orqali)	125–250	**200 × 150**	7.5
XZ1 (o'rta devor yonida)	80	**200 × 150**	5.9
magistral: 6-xona → XZ2 → XZ3 → 2-xona (markazdan)	1 080	**400 × 200**	10.5
magistral: 6-xona → XZ2 → XZ3 → 2-xona (markazdan)	1 210	**400 × 250**	1.2
magistral: 6-xona → XZ2 → XZ3 → 2-xona (markazdan)	1 650	**500 × 250**	10.7
magistral: 6-xona → XZ2 → XZ3 → 2-xona (markazdan)	2 690	**500 × 300**	0.3

VV1: kanalli ventilyator 3 000 m³/soat, ~400 Pa (to'rtburchak, 500×300 ga mos), shovqin so'ndirgich bilan; kanallar jami ~73 m.

O'rnatish

- Kanallar shift ostida, osmalarda (har 2–2.5 m); pastki qirrasi ≥ 2.9 m. Shisha devorlar ustidagi GKL (2.5–3.5 m) va mavjud devorlardan teshik orqali o'tadi; teshik atrofi germetiklanadi.
- Ventilyator osma rezina amortizatorlarda, kanalga egiluvchan ulanadi; har liniya boshida rostlash klapani, panjaralar sozlanadigan (havo — jadval bo'yicha).
- Kanallar 2-varaqdagi konditsioner kanallari bilan 2 joyda kesishadi — so'rish kanali yuqorida, berish kanali ostida (2-varaq, ①–②).

Diqqat — chiqish joyi: VV1 va chiqish — 2-xona fasadida (19.10), konditsionerning tashqi havo olish panjarasi — 3-xona fasadining sharqiy burchagida: oralig'i ~8 550 mm (me'yor ≥ 8 m — mos). Chiqish oyna ustidagi ~400 mm devordan (panjara ~500 × 300); joy yetmasa — tomga (tom sathidan ≥ 1 m baland, soyabon bilan).

PDP Academy · Xadra filiali	
Chizma	**Vitechka — so'rish kanallari rejasi**
Ish chizmasi — ustalar uchun. Buyurtmachi eskizi (15.10) va qarorlari (16.10) asosida. Toshkent sh., Xadra — 2-qavat	
Masshtab: 1:100	Sana: 16.10.2026
Versiya: v1.2	Varaq: 1 / 2
Bajardi: _____	Tekshirdi: _____

2-varaq · Konditsioner (kanalli) va tashqi havo — kanallar rejasi, 1:100



KANALLI KONDITSIONER — BERISH KANALLARI (16.10)

- sovuq havo kanali (kesim × m³/soat)
- tashqi havo kanali (30%) — sharqiy devor bo'ylab bloklar orqasiga; 3- va 4-xonada GKL qutida
- vitechka (1-varaq, xira)
- shift diffuzori D… (4 tomonga)
- KK — ichki blok, kladovka shifti ustida
- ①–② kesishuvlar

Ichki bloklar — kladovka (KL1, KL2) shifti ustida, orqasi tashqi havo kanali tomonda (96 000 BTU deb olindi)

Blok	Xonalar	Havo	Tashqi havo
KK1	1, 2, 3, 4 (1 diffuzor), XZ3 · ustozlar, XZ4 · call-markaz	5 000	1 500 (30%)
KK2	5, 6, 7, 4 (1 diffuzor), XZ1 · sotuv, XZ2 · admin	5 000	1 500 (30%)

Quvvat tekshiruvi (kW, eng issiq payt)

Blok	Xonalar	+ tashqi havo 30%	Jami	96k (24.4 kW issiqda)
KK1	22.9–28.2	7.0	**35.2**	yetmaydi (−10.8 kW)
KK2	25.5–31.1	7.0	**38.2**	yetmaydi (−13.7 kW)

Xonalar: havo, tashqi havo, so'rish, CO_2

Xona	Odam	Havo	Tashqi (30%)	So'rish	CO_2, ppm
1-xona (KK1)	25	1 180	354	320	1 690
2-xona (KK1)	25	1 320	396	360	1 560
3-xona (KK1)	25	1 320	396	360	1 560
4-xona (KK1 + KK2)	19	920	276	250	1 660
XZ4 · call-markaz (KK1)	5	310	93	80	1 390
XZ3 · ustozlar (KK1)	9	420	126	110	1 710
5-xona (KK2)	25	1 340	402	360	1 540
6-xona (KK2)	25	1 340	402	360	1 540
7-xona (KK2)	25	1 340	402	360	1 540
XZ1 · sotuv (KK2)	6	300	90	80	1 620
XZ2 · admin (KK2)	4	200	60	50	1 620

Diffuzorlar — 18 ta

Kod	Xona	x	y	m³/soat	Panel	Patrubok
D1	1-xona	8 150	6 200	590	600×600	250×200
D2	1-xona	10 050	6 200	590	600×600	250×200
D3	2-xona	14 100	6 200	660	600×600	300×200
D4	2-xona	16 000	6 200	660	600×600	300×200
D5	3-xona	19 650	6 200	660	600×600	300×200
D6	3-xona	22 300	6 200	660	600×600	300×200
D7	4-xona	20 900	10 900	450	450×450	250×150
D8	4-xona	20 900	13 700	470	450×450	250×150
D9	5-xona	19 650	18 200	670	600×600	300×200
D10	5-xona	22 300	18 200	670	600×600	300×200
D11	6-xona	14 100	18 200	670	600×600	300×200
D12	6-xona	16 000	18 200	670	600×600	300×200
D13	7-xona	8 100	18 200	670	600×600	300×200
D14	7-xona	10 000	18 200	670	600×600	300×200
D15	XZ4 · call-markaz	10 750	10 600	310	450×450	200×150
D16	XZ3 · ustozlar	15 500	10 600	420	450×450	250×150
D17	XZ1 · sotuv	10 750	13 800	300	450×450	200×150
D18	XZ2 · admin	16 300	13 800	200	450×450	200×150

Magistral	m³/soat	Kesim, mm	Uzunlik, m
K1 koridori (KK1 dan)	590	**250 × 200**	1.9
K1 koridori (KK1 dan)	1 180	**400 × 250**	0.7
K1 koridori (KK1 dan)	1 490–2 150	**500 × 250**	4.8
K1 koridori (KK1 dan)	2 570	**500 × 300**	0.5
K1 koridori (KK1 dan)	3 230	**600 × 300**	3.6
K1 koridori (KK1 dan)	3 890	**600 × 400**	1.1
K1 koridori (KK1 dan)	4 550–5 000	**700 × 400**	0.8
K2 koridori (KK2 dan)	670	**250 × 200**	1.9
K2 koridori (KK2 dan)	1 340	**400 × 250**	0.8
K2 koridori (KK2 dan)	1 640	**500 × 250**	3.4
K2 koridori (KK2 dan)	2 310	**500 × 300**	1.9
K2 koridori (KK2 dan)	2 980–3 180	**600 × 300**	3.6
K2 koridori (KK2 dan)	3 850	**600 × 400**	1.1
K2 koridori (KK2 dan)	4 520–4 990	**700 × 400**	0.8
tashqi havo → KK1 (yon ulanish)	1 500	**500 × 250**	0.3
tashqi havo: 3-xona fasadi → sharqiy devor bo'ylab → KK2	1 500	**500 × 250**	7.6
tashqi havo: 3-xona fasadi → sharqiy devor bo'ylab → KK2	3 000	**600 × 300**	9.0

Kanallar jami: berish ~74 m (patrubokli), tashqi havo ~17 m, vitechka ~73 m — hammasi ~163 m (narx: 300 000 so'm/m). Qaytish havosi (QP): koridor shiftidagi panjara 600 × 300, kladovka devori yonida (logoli variantda reykalar ortida) — shift ustidagi bo'shliq orqali blokning qaytish kamerasiga.

Diqqat: (1) 2 × 96k yetmaydi — har blokka ~38 kW (~130k BTU) kerak; 96k issiqda ~24.4 kW beradi. (2) 30% tashqi havoda to'la sinfda CO_2 ~1 400–1 700 ppm (me'yor ≤ 1 000–1 200): darsdan keyin shamollatish yoki tashqi havoni ~45–50% ga oshirish (blok shunga mos tanlansa) tavsiya.

PDP Academy · Xadra filiali

Chizma	**Konditsioner (kanalli) va tashqi havo**

Ish chizmasi — ustalar uchun. Buyurtmachi eskizi (15.10) va qarorlari (16.10) asosida. Toshkent sh., Xadra — 2-qavat

Masshtab	Sana	Versiya	Varaq
1:100	16.10.2026	v1.2	2 / 2

Bajardi: _____ Tekshirdi: _____